Compare deployment models

Task 1

Pick a use case from an app you use daily (for example: Instagram photo sharing, Zomato food ordering, or Flipkart online shopping) and briefly describe the main technical requirements for hosting this app online.

Use case chosen: Zomato (food ordering app)

Hosting an app like Zomato online requires several key technical components working together. First, it needs web and application servers to handle user requests such as browsing restaurants, placing orders, and tracking deliveries. Second, it needs a reliable database to store restaurant listings, menus, user accounts, and order history. Third, it needs object storage for images such as restaurant photos and food items. It also needs APIs for third-party integrations like payment gateways, maps for delivery tracking, and SMS/push notifications. Since food ordering apps experience heavy traffic during lunch and dinner hours, the infrastructure must support load balancing and auto-scaling to handle sudden spikes without slowing down. Finally, strong security measures are required to protect sensitive user data such as addresses and payment details.

Task 2

Create a table comparing public, private, and hybrid cloud deployment models for your chosen use case. Include at least 4 factors: cost, security, scalability, and ease of maintenance.

Factor	Public Cloud	Private Cloud	Hybrid Cloud
Cost	Low upfront cost; pay-as-you-go, ideal for a growing app like Zomato.	High upfront cost since Zomato would need to own and maintain dedicated servers.	Moderate cost; only critical/sensitive workloads run on costlier private infrastructure, rest on public cloud.
Security	Good security, but data is shared on provider's infrastructure with other tenants.	Highest security and control, best suited for storing sensitive payment and user data.	Strong security for sensitive data (kept private) while less sensitive data uses public cloud.
Scalability	Excellent scalability; can instantly add resources during lunch/dinner rush or big sale events.	Limited scalability since it depends on physically owned hardware capacity.	Very good scalability; public cloud portion absorbs sudden traffic spikes while private handles core data.
Ease of Maintenance	Easiest to maintain since the provider manages hardware, patching, and updates.	Hardest to maintain; Zomato's own team must manage hardware, security patches, and updates.	Moderate effort; public portion is managed by the provider, private portion still needs an in-house team.

Task 3

Write a short paragraph explaining which cloud deployment model you would recommend for your chosen use case and why. Support your answer with at least two points from your comparison table.

For an app like Zomato, a hybrid cloud model would be the most suitable choice. As shown in the comparison table, the hybrid model offers very good scalability, allowing the public cloud portion to instantly absorb sudden traffic spikes during lunch and dinner rush hours or big sale events, without Zomato needing to over-invest in permanent private hardware. At the same time, it provides strong security for sensitive information such as customer payment details and personal data, since that portion of the workload can be kept on a private, tightly controlled environment. This balance of flexibility, cost efficiency, and control makes hybrid cloud a practical fit for a fast-growing app that handles both public-facing traffic and sensitive user data.

Task 4

Imagine your chosen app becomes extremely popular overnight (like during IPL finals or a big sale). Describe one advantage and one disadvantage of using a hybrid cloud model in this situation.

Advantage:
If Zomato suddenly sees a massive spike in orders overnight, for example during an IPL final when everyone orders food at once, the public cloud portion of the hybrid setup can automatically scale up and add extra servers within minutes. This means the app can keep functioning smoothly and handle millions of extra requests without crashing, while the private cloud continues to securely manage sensitive user and payment data.

Disadvantage:
Managing a hybrid cloud during such a sudden surge can become complex, since data and traffic constantly need to move between the private and public environments. If this connection between the two environments is not perfectly configured, it can create delays, synchronisation issues, or even temporary inconsistencies in order data, making the hybrid setup harder to manage smoothly compared to a purely public cloud during extreme, unpredictable spikes.

Task 5

Use ChatGPT or any AI tool to generate a summary of when to use public, private, and hybrid clouds for popular Indian apps (such as Paytm, IRCTC, BookMyShow). Paste the summary and highlight one insight you found surprising.

AI-generated summary:

> *Public cloud works best for apps that need to scale quickly and cost-effectively for a large number of general users, such as BookMyShow during ticket releases, since it can handle unpredictable spikes without heavy upfront investment. Private cloud is preferred when an application handles highly sensitive data under strict regulation, such as Paytm's core payment and banking operations, where full control over infrastructure and compliance is essential. Hybrid cloud is often the ideal middle ground for apps like IRCTC, which need to keep critical passenger and payment data secure on private infrastructure, while still using public cloud resources to manage the massive, unpredictable traffic surges that happen during festival season ticket bookings, keeping the system both secure and highly available.*

Insight I found surprising:

The most surprising insight was that even a government-linked platform like IRCTC, which people often assume runs entirely on private, fully-owned servers for maximum security, actually benefits more from a hybrid approach. It was interesting to realise that relying only on private cloud would likely make IRCTC's website crash even more often during Tatkal booking rushes, since private infrastructure alone cannot scale as fast as public cloud resources can during those extreme short bursts of demand.